Financial Management

By:
Dr. Vaishali Pagaria

vaishali.pagaria@wilp.bits-pilani.ac.in

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Module 2: Risk Management Part - 1

Risk and Return

- Return is the expected cash inflows from the investment
- Risk is associated with the uncertain cashflow
- Risk needs to be assessed and incorporated in the decision making

Calculating Return

Income received on an investment plus any change in market price, usually expressed as a percent of the beginning market price of the investment.

$$\textbf{Simple Return (R)} = \frac{D_t + (P_t - P_{t-1})}{P_{t-1}}$$

Where,

D_t = Dividend paid

P_t = Price of stock at time *t*

P_{t-1} = Price of stock at time *t-1*

Geometric Vs. Arithmetic Average Rate of Return

Arithmetic Mean = $\frac{\sum(x_1 + x_2 + x_3 + \ldots\ldots\ldots + x_n)}{n}$

Arithmetic Rate of Return = $\frac{\sum(R_1 + R_2 + R_3 + \ldots\ldots\ldots + R_n)}{n}$

The arithmetic average rate of return answers the question, "what was the average of the yearly rates of return?

Geometric Vs. Arithmetic Average Rate of Return

Geometric Mean = $\sqrt[n]{x_1 * x_2 * x_3 * \ldots\ldots\ldots * x_n}$

Geometric Rate of Return = $\sqrt[n]{R_1 * R_2 * R_3 * \ldots\ldots\ldots * R_n}$

The geometric average rate of return answers the question, “What was the growth rate of your investment?” (CAGR)

Adjustment in Geometric Rate of Return is required to calculate negative return.

Arithmetic average may not always capture the true rate of return realized on an investment. In some cases, geometric or compound average may be a more appropriate measure of return

Risk

- The variability of returns from those that are expected.
- Once the risk characterising future cashflows is measured, an appropriate risk-adjusted discount rate should be applied to convert future cash flows into their present values.
- To estimate the risk-adjusted discount rate – one has to measure and price financial risk.

Expected Return (Discrete Distribution)

$\bar{R} = \sum_{i=1}^{n} (R_i)(P_i)$

Where,

$\bar{R}$ is the expected return for the asset,

R_i is the return for the ith possibility,

P_i is the probability of that return occurring,

n is the total number of possibilities.

Calculation of Risk (Discrete Distribution)

$$\text{Risk } (\sigma) = \sqrt{\sum_{i=1}^{n} (R_i - \overline{R})^2 \; X \; (P_i)}$$

Standard Deviation, σ, is a statistical measure of the variability of a distribution around its mean.

It is the square root of variance.

Note, this is for a discrete distribution

Illustration

Calculate expected return and risk of the following stock.

Time	Return	Probability
1	10%	0.1
2	11%	0.2
3	10.5%	0.4
4	12%	0.2
5	11%	0.1

Expected Return (Continuous Distribution)

$$\overline{R} = \sum_{i=1}^{n} (R_i) /(n)$$

Where,

$\overline{R}$ is the expected return for the asset,

R_i is the return for the ith observation,

n is the total number of observations

Calculation of Risk (Continuous Distribution)

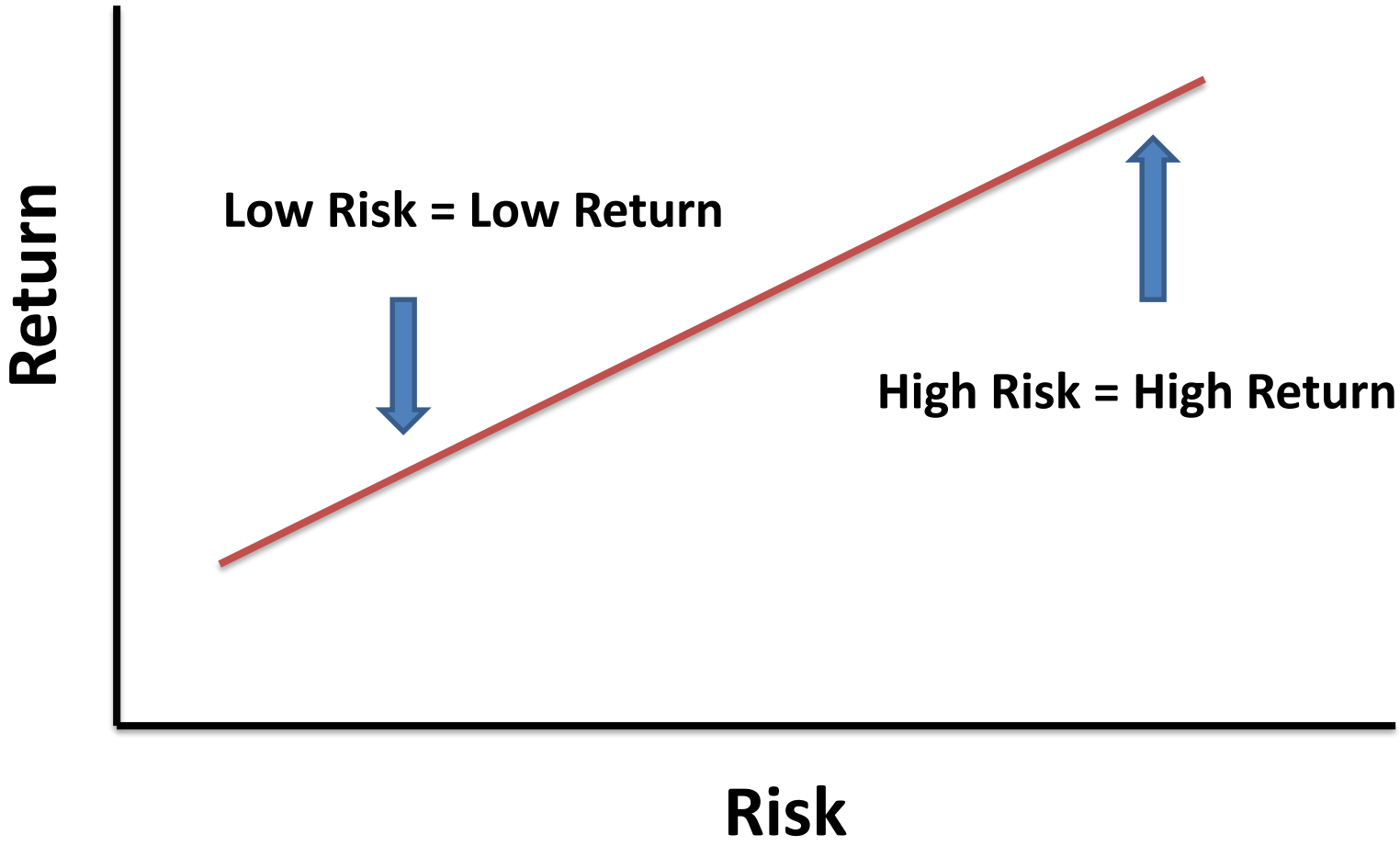
$$\text{Risk } (\sigma) = \sqrt{\frac{\sum_{i=1}^{n}(R_i - R)^2}{n}}$$

Note, this is for a continuous distribution where the distribution is for a *population*

Risk and Return Relationship

- The correlation between the chance one take to invest in the business and the performance of the business states the relation between risk and return.
- Risk-Return Trade Off says that higher the risk, higher is the return….*though it is not guaranteed*.
- Securities are risky because their returns are variable.
- The most commonly used measure of risk or variability in finance is ***standard deviation***.
- The risk of a security can be split into two parts: unique risk and market risk
- Unique risk stems from firm-specific, whereas market risk emanates from economy-wide factors.

Risk- Return Trade Off



Risk and Return Relationship

Required rate of return = Risk-free rate of return + Risk premium

Risk-free rate of return (Rf) refers to the return available on a security with no risk of default.

Risk-free rate of return = Real Rate of Return + Expected Inflation Premium

A *risk premium* is a potential “reward” that an investor expects to receive when making a risky investment.

Portfolio Theory

- Combination of different investment in one basket
- Proportion is decided based on the risk-return relationship and the risk attitude
- Risk management through diversification of portfolio

Co-efficient Variance

- Coefficient of Variation is a measure of relative dispersion (risk) or a measure of risk per unit of expected return.
- It converts standard deviation of expected values into relative values to enable comparison of risks associated with assets having different expected values

$$CV = \frac{\sigma}{Expected\ Return\ (\overline{R})}$$

- The larger the CV, the larger the relative risk of the asset. As a rule, the use of the coefficient of variation for comparing asset risk is the best since it considers the relative size (expected value) of assets.

Correlation Coefficient

- Correlation coefficient is a measure of the degree of correlation between two series
- For given weights, portfolio standard deviation declines as correlation coefficient moves from + 1.0 to – 1.0

$$\rho_{12} = \frac{CV_{12}}{\sigma_1 \, \sigma_2}$$

Determining Portfolio Expected Return (Two Securities)

$$\bar{R}_p = \sum_{i=1}^{n} (W_i)(R_i)$$

Where,

$\bar{R}_p$ is the expected return for the asset,

W_i is the weight (investment proportion) for the ith asset in the portfolio,

R_i is the expected return of the ith asset,

n is the total number of assets in the portfolio

Determining Portfolio Risk (Two Securities)

$$\text{Var(Rp)} = W_1^2 \text{ Var } (R_1) + W_2^2 \text{ Var } (R_2) + 2W_1 \, W_2 \text{ Cov } (R_1, R_2)$$

Or

$$\sigma_P = Squroot\ of\ \sigma_P^2 = W_1^2\sigma_1^2 + W_2^2\ \sigma_2^2 + 2W_1\ W_2\rho_{12}\sigma_1\ \sigma_2$$

$$\sigma_P = \sqrt{\sum_{j=1}^{m}\sum_{k=1}^{m} W_i\, W_k\ \sigma_{jk}}$$

Where,
W_j is the weight (investment proportion) for the jth asset in the portfolio,
W_k is the weight (investment proportion) for the kth asset in the portfolio,
σ_{jk} is the covariance between returns for the jth and kth assets in the portfolio
or
can use Risk $(\sigma) = \sqrt{\sum_{i=1}^{n}(R_i - R)^2\ X\ (P_i)}$

Illustration: Portfolio Expected Return and Risk

State of the Economy	Probability	Return on Stock A	Return on Stock B	Return on Portfolio
1	0.2	15%	-5%	5%
2	0.2	-5%	15%	5%
3	0.2	5%	25%	15%
4	0.2	35%	5%	20%
5	0.2	25%	35%	30%

(a) What is the expected return and risk if you invest only in stock A?
(b) What is the expected return and risk if you invest only in stock B?
(c) What is the expected return and risk if you invest in a portfolio consisting of stock A and B in equal proportion?

Practice Problem

	Economic Condition			
	High growth	Low growth	Stagnation	Recession
Probability	0.3	0.4	0.2	0.1
Return on Bindra's stock (%)	10	11	12	14
Return on Cinox's stock (%)	15	13	9	6

(a) What is the expected return and risk if you invest only in Bindra's stock?
(b) What is the expected return and risk if you invest only in stock Cinox' stock?
(c) What is the expected return and risk if you invest in a portfolio consisting of Bindra and Cinox in equal proportion?

Practice Problem: Calculate Portfolio Return and Risk

Year	Airtel	Mahindra & Mahindra
2019	25%	-4%
2020	-8%	28%
2021	10%	-6%
2022	11%	48%
2023	36%	23%

Equal weightage of both the securities in the portfolio

CAPITAL ASSET PRICING MODEL (CAPM)

- The capital asset pricing model (CAPM), as the name suggests, is a theory that explains how asset prices are formed in the market place. It is a logical and major extension of the portfolio theory of Markowitz by William Sharpe (1964), John Lintner (1965) and Jan Mossin (1967).
- The capital asset pricing model provides the framework for determining the equilibrium expected return for risky assets.
- It uses the results of capital market theory to derive the relationship between expected return and systematic risk of individual assets/securities and portfolios.

Implications of CAPM

(a) Risk-return relationship for an efficient portfolio

(b) Risk-return relationship for an individual asset/security

(c) Identification of under- and over- valued assets traded in the market

(d) Pricing of assets not yet traded in the market

(e) Effect of leverage on cost of equity (rate of return required by equity shareholders)

(f) Capital budgeting decisions and cost of capital

(g) Risk of the firm through diversification of project portfolio

Assumptions of CAPM

- All investors are price-takers. Their number is so large that no single investor can affect prices.
- All investors use the mean-variance portfolio selection model of Markowitz.
- Assets/securities are perfectly divisible.
- All investors plan for one identical holding period.
- Homogeneity of expectation for all investors results in identical efficient frontier and optimal portfolio.
- Investors can lend or borrow at an identical risk-free rate.
- There are no transaction costs and income taxes

Elements of the Model

CAPM Approach: calculates the expected return on a security based on its level of risk.

$$Expected\ return\ on\ equity = k_e = r_f + \beta\,(r_m - r_f)$$

Where,

r_f **= risk free rate :** this may be estimates as the yield on a long-term government bond that has a maturity of 10 years or more

β = coefficient of equity: beta is the measure of risk involved with investing in a particular stock relative to the risk of the market. This may be calculated by regressing the monthly returns on the stock over the monthly returns on the market index over the past 60 months or more.

r_m **= return on the market:** this may be estimated by a broad stock market index like the BSE National Index, NSE

$r_m - r_f$ **= market risk premium:** this may be estimated as the difference between average return on the market portfolio and the average risk free rate over the past 10 to 30 years – the longer the period, the better it is.

Risk Attitudes

Certainty Equivalent (CE) is the amount of cash someone would require with certainty at a point in time to make the individual indifferent between that certain amount and an amount expected to be received with risk at the same point in time

Risk Attitudes

Risk Preference	• Certainty equivalent > Expected value
Risk Neutral	• Certainty equivalent = Expected value
Risk Aversion	• Certainty equivalent < Expected value

Most individuals are Risk Averse

Impact of Inflation on Return

- Investment is made with the aim of increasing the future purchasing power.
- Understanding the rate of inflation is important because it represents the rate at which the real value of an investment is eroded and therefore loss in purchasing power over a period of time.
- Inflation also tells investors exactly how much of a return their investments need to make for them to maintain their standard of living.
- Inflation is ultimately about money growth, and it is a reflection of too much money chasing too few products.
- Purchasing power of each money declines.
- Therefore, the investor should try to invest in an investment with return that are equal to or greater than inflation.

THANK YOU!!!